

improved sensitivity, and increased automation. The potential impact of MEMS and microfluidics fused devices is immense. They have the potential to revolutionise healthcare by enabling point-of-care diagnostics, personalised medicine, and portable lab-on-a-chip platforms. Moreover, they can find applications in environmental monitoring, food safety, and industrial processes.

Cancer diagnostics using MEMS and microfluidics

MEMS sensors have shown promise in the field of cancer detection and identification. These sensors are miniature devices that integrate mechanical and electrical components on a microscopic scale. When it comes to cancer cell identification, MEMS sensors offer several advantages such as:

Size and sensitivity. MEMS sensors are extremely small, allowing them to be easily integrated into lab-on-a-chip devices or handheld diagnostic tools. Their small size enables them to interact with individual cells or small clusters of cells. These sensors can detect subtle changes in physical properties such as size, shape, and mechanical behaviour, which are indicative of cancer cells.

Label-free detection. One significant advantage of MEMS sensors is their ability to detect cancer cells without the need for specific labels or markers. This label-free detection approach eliminates the requirement for additional processing steps, reducing the complexity and time required for analysis. MEMS sensors can directly interact with cells and provide real-time, label-free information.

Mechanical characterisation. Cancer cells exhibit distinct mechanical properties compared to healthy cells. MEMS sensors can measure mechanical properties such as stiffness, deformability, and viscoelasticity of cells. By subjecting cells to

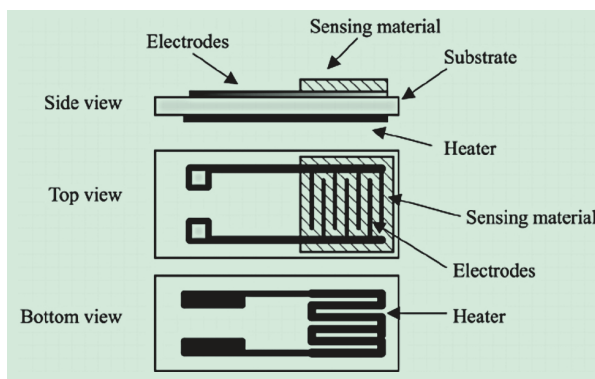


Fig. 2: Structure of a MEMS sensor

controlled mechanical forces, MEMS sensors can detect differences in cell behaviour, identifying cancer cells based on their unique mechanical signatures.

High throughput screening.

MEMS sensors can be designed to operate in an array format, allowing for high-throughput screening of cells. Multiple sensors can be fabricated on a single chip, enabling simultaneous analysis of multiple cells or cell populations. This feature is particularly useful for applications where a large number of cells need to be analysed rapidly such as in cancer diagnostics or drug screening.

Real-time monitoring. MEMS sensors offer the potential for real-time monitoring of cancer cells. By continuously monitoring cellular behaviour and changes in physical properties, these sensors can provide dynamic information about cell growth, migration, and response to treatment. Real-time monitoring allows for early detection of cancer cells and facilitates the evaluation of treatment efficacy.

In conclusion, MEMS sensors provide a promising approach for the identification of cancer cells. Their small size, sensitivity, label-free detection, ability to characterise mechanical properties, high throughput screening capability, and potential for real-time monitoring make them valuable tools in cancer diagnostics and research. Continued advance-

ments in MEMS sensor technology hold the potential to revolutionise cancer detection and improve patient outcomes through early diagnosis and personalised treatment strategies.

Applications

MEMS sensors, or Micro-Electro-

Mechanical Systems sensors, have shown promising potential in various fields, including cancer diagnostics. Their small size, high sensitivity, and ability to integrate with microelectronics make them ideal for developing advanced diagnostic devices. Here, we explore the applications of MEMS sensors in cancer diagnostics devices.

1. Biomarker detection. MEMS sensors can be utilised to detect specific biomarkers associated with cancer. These sensors can detect minute changes in the concentration of biomarkers present in bodily fluids such as blood or urine. By accurately measuring these biomarkers, MEMS sensors enable early detection of cancer, monitoring disease progression, and assessing treatment efficacy.

2. Liquid biopsies. MEMS sensors are being integrated into liquid biopsy devices, which are non-invasive alternatives to traditional tissue biopsies. Liquid biopsies analyse circulating tumour cells (CTCs) and cell-free DNA (cfDNA) in the blood. MEMS sensors play a crucial role in capturing, isolating, and analysing these cancer-related cells and genetic materials, offering valuable insights into tumour presence, mutation profiling, and treatment response.

3. Point-of-care testing. MEMS-based cancer diagnostic devices can facilitate point-of-care testing, bringing diagnostic capabilities closer to